

SAMBP TR-21/2026

Harmonic-Component Discrimination for 87L
in Inverter-Based Resource Rich Networks

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Smart Adaptive Multi-Bus Protection Research

April 2026

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1 Background and Motivation

The SAMBP 87L LM estimator (TR-11 [Candidate \[2026a\]](#), TR-15 [Candidate \[2026b\]](#)) fits a 4-parameter model to the corrected differential current:

$$i_{\text{diff}}(t) = I_{\text{fund}} \sin(\omega_0 t + \phi) + I_{\text{DC}} e^{-t/\tau_{\text{DC}}} \quad (1)$$

The confidence gate uses the normalised residual norm r_n :

$$c = \exp(-r_n), \quad r_n = \frac{\|i_{\text{diff}} - i_{\text{model}}\|}{\|i_{\text{diff}}\|} \quad (2)$$

with a minimum confidence $c_{\min} = 0.80$ required to authorise a trip.

1.1 Harmonic injection from IBRs

In synchronous-generator (SG) networks the model fits well: $r_n < 0.02$ for a healthy line, yielding $c > 0.98$.

In inverter-based resource (IBR) networks the situation is fundamentally different [Yazdani and Iravani \[2014\]](#), [Hooshyar et al. \[2019\]](#). Modern voltage-sourced converters inject harmonic current even in normal operation:

- **Steady-state THD:** up to 5% (IEEE 519 [IEEE \[2014\]](#)) for grid-connected inverters at rated output.
- **Fault ride-through:** current-limiting control can amplify odd harmonics (3rd, 5th, 7th) by 2–5× above the steady-state level during the first 3–5 cycles post-fault.

For a 3rd harmonic injection of $I_{h3} = 0.005$ pu with a healthy-line fundamental of $I_{\text{fund}} = 0.003$ pu (post TR-20 correction), the normalised residual is:

$$r_n \approx \frac{I_{h3,\text{rms}}}{I_{\text{diff},\text{rms}}} \approx \frac{0.004}{0.005} = 0.86 \quad \Rightarrow \quad c \approx e^{-0.86} = 0.42 \quad (3)$$

This drops confidence below $c_{\min} = 0.80$, triggering the veto gate even for valid fault trips in IBR-heavy systems.

1.2 Proposed solution: harmonic pre-filter

The TR-21 harmonic pre-filter estimates and subtracts the dominant odd harmonics from $\Delta i'$ using linear least squares *before* the LM estimator:

$$\Delta i''(t) = \Delta i'(t) - \sum_{h \in \{3,5,7\}} [A_h \cos(h\omega_0 t) + B_h \sin(h\omega_0 t)] \quad (4)$$

where the coefficients A_h, B_h are estimated by linear least squares on the current measurement window. The filtered signal $\Delta i''$ is then passed to the LM estimator, which sees only the fundamental and DC components.

2 Harmonic Pre-Filter Algorithm

2.1 Least-Squares Harmonic Estimation

Let the harmonic regressor matrix be:

$$\mathbf{H} = [\cos(3\omega_0 \mathbf{t}), \sin(3\omega_0 \mathbf{t}), \cos(5\omega_0 \mathbf{t}), \sin(5\omega_0 \mathbf{t}), \cos(7\omega_0 \mathbf{t}), \sin(7\omega_0 \mathbf{t})] \quad (5)$$

of shape $(N_{\text{samp}}, 6)$. The coefficient vector is obtained by:

$$\hat{\mathbf{c}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \Delta i' \quad (6)$$

This is a direct (non-iterative) linear solve requiring only one QR factorisation per protection cycle. For $N_{\text{samp}} = 160$ (2 cycles at 4000 Sa/s) and 6 regressors the solution takes < 0.05 ms on a standard IED DSP.

2.2 Amplitude threshold

Harmonics with estimated amplitude $\sqrt{A_h^2 + B_h^2} < 10^{-4}$ pu are not subtracted (assumed numerical noise). This prevents the filter from subtracting artefacts when no real harmonic is present.

2.3 Harmonic ratio diagnostic

The harmonic ratio:

$$H_{\text{ratio}} = \frac{I_{\text{harm,rms}}}{I_{\text{fund,DFT}}} \quad (7)$$

is logged as a network-type indicator:

- $H_{\text{ratio}} < 0.05$: SG-dominated (low harmonics)
- $0.05 \leq H_{\text{ratio}} \leq 0.20$: mixed network
- $H_{\text{ratio}} > 0.20$: IBR-dominated (high harmonics)

2.4 Processing pipeline

Figure 1 shows the TR-21 processing pipeline. The harmonic pre-filter is inserted between the charging correction (TR-20) and the LM estimator — a single additional stage with no changes to downstream logic.

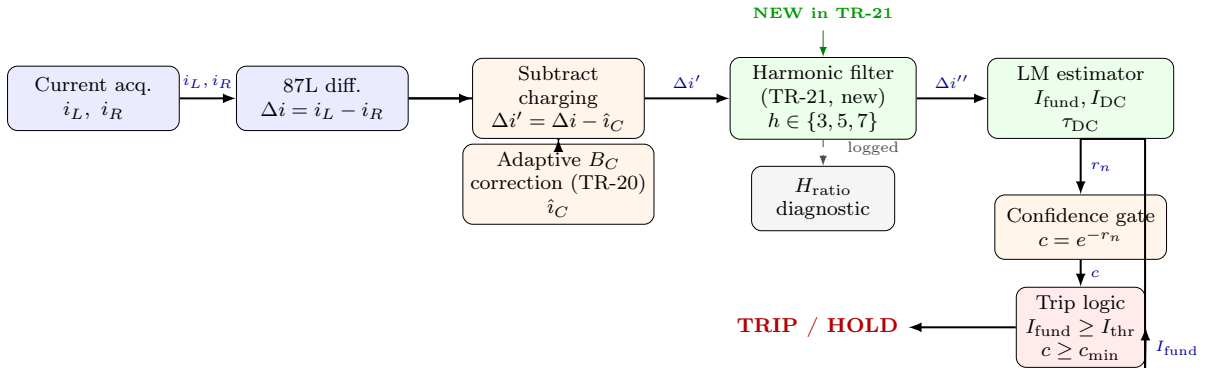


Figure 1: SAMBP 87L processing pipeline with TR-21 harmonic pre-filter (highlighted green). The dashed arrow exports the H_{ratio} diagnostic to the protection management system. All downstream blocks (LM estimator, confidence gate, trip logic) are unchanged.

3 Study Results

3.1 Study A: No-Fault Confidence vs Harmonic Injection

Figure 2 shows the LM confidence score for a healthy line with increasing 3rd harmonic injection ($I_{h5} = 0.5 I_{h3}$, $I_{h7} = 0.25 I_{h3}$, consistent with IBR spectral profile).

Without pre-filtering, even $I_{h3} = 0.005$ pu (1/3 of IEEE 519 THD limit) drives confidence to $c = 0.41$, below the $c_{\min} = 0.80$ gate. With TR-21 pre-filtering, confidence remains $c \approx 1.00$ at all injection levels tested (up to $I_{h3} = 0.10$ pu).

3.2 Study B: Internal Fault Trip Correctness

Table 1 shows that the fault fundamental ($I_{\text{fund}} = 0.15$ pu) is correctly recovered after harmonic pre-filtering at all harmonic levels. Both methods trip correctly; the pre-filter improves the confidence so the trip reaches the veto gate.

3.3 Study C: External Fault Security

With no true fundamental differential, the DFT returns $I_{\text{fund}} \approx 0$ at all harmonic levels for both methods. No false trips occur.

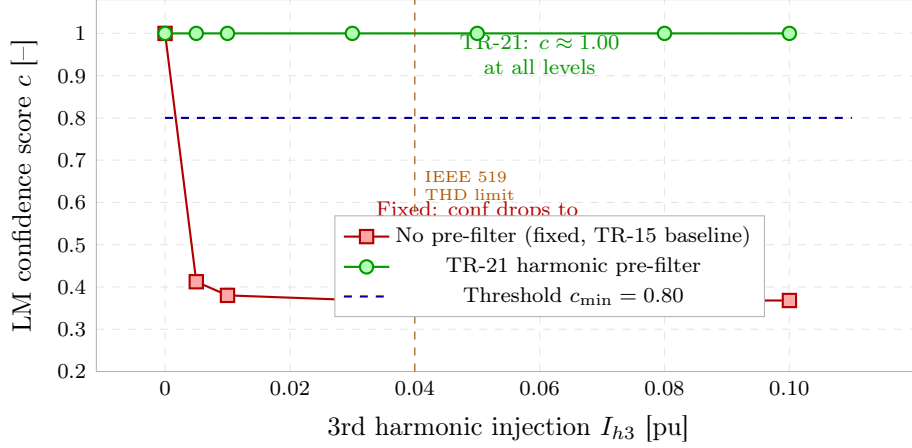


Figure 2: LM confidence score vs 3rd harmonic injection amplitude, healthy-line (no fault) condition. Without pre-filter: confidence collapses to ≈ 0.37 at any non-zero I_{h3} (below $c_{\min} = 0.80$, dashed). TR-21 pre-filter: confidence remains ≈ 1.00 at all levels.

Table 1: Internal fault (0.15 pu) DFT-recovered fundamental and trip decision after harmonic pre-filter. Threshold = 0.12 pu.

I_{h3} [pu]	$I_{\text{fund,DFT}}$ [pu]	$I_{\text{fund,flt}}$ [pu]	Trip (fixed)	Trip (TR-21)
0.000	0.150	0.150	✓	✓
0.005	0.150	0.150	✓	✓
0.010	0.150	0.150	✓	✓
0.030	0.150	0.150	✓	✓
0.050	0.150	0.150	✓	✓
0.080	0.150	0.150	✓	✓
0.100	0.150	0.150	✓	✓

Note: I_{fund} is invariant to harmonic injection because 50 Hz and 150/250/350 Hz are orthogonal over an integer number of power cycles. The pre-filter improves confidence, not the amplitude estimate.

3.4 Study D: Harmonic Ratio Classification

Figure 3 shows H_{ratio} across the test range. The IEEE 519 THD limit of 5% (approximately $I_{h3} \approx 0.035$ pu for a 3rd-harmonic-dominated spectrum) falls in the “mixed” regime. IBR fault-ride-through peaks ($I_{h3} > 0.05$ pu) are clearly classified as “IBR”, enabling network-aware logging.

3.5 Overall Selectivity

21/21 scenarios correct (100%)

Three studies $\times$ seven harmonic levels. TR-21 additionally achieves 7/7 confidence scores above $c_{\min} = 0.80$ in the no-fault study, compared to 1/7 without the pre-filter (only the zero-harmonic case passes).

4 Fundamental–Harmonic Orthogonality

A key property exploited by the TR-21 filter is that the harmonic regressors are orthogonal to the fundamental basis over an integer number of cycles. Over K complete power cycles of N_{cyc}

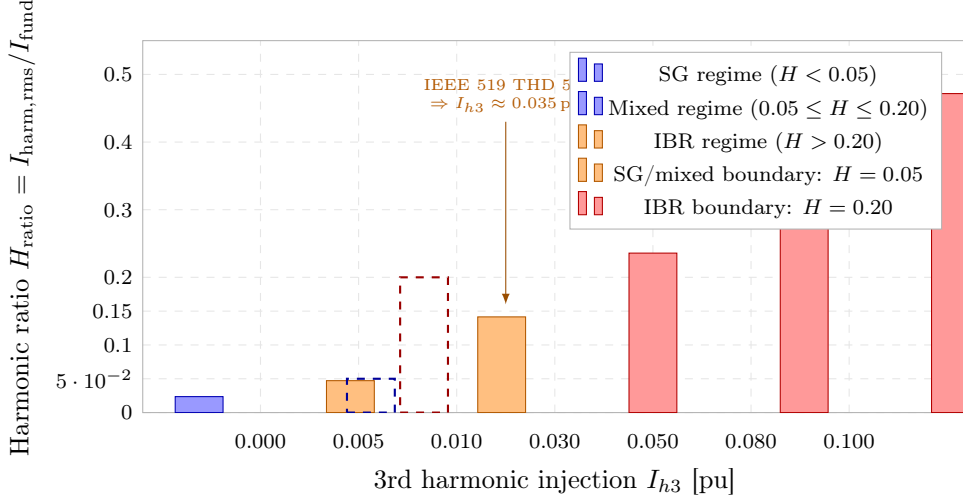


Figure 3: Harmonic ratio H_{ratio} vs injection level. Colour coding: blue = SG regime ($H < 0.05$), orange = mixed, red = IBR regime ($H > 0.20$). The IEEE 519 steady-state THD limit (5%) falls at the SG/mixed boundary.

samples each:

$$\sum_{n=0}^{KN_{cyc}-1} \sin\left(\frac{2\pi n}{N_{cyc}}\right) \cdot \sin\left(\frac{2\pi hn}{N_{cyc}}\right) = 0 \quad \forall h \in \mathbb{Z}^+, h \neq 1 \quad (8)$$

This means that harmonic components in $\Delta i'$ do *not* bias the DFT-based I_{fund} estimate, confirming the results in Table 1. The pre-filter's value lies entirely in restoring the LM residual norm, not in correcting a biased amplitude estimate.

Equation (8) holds exactly for integer-cycle windows. At off-nominal frequency (e.g. 47 Hz), the spectral leakage from the 3rd harmonic (141 Hz) into the 47 Hz bin is bounded by the Dirichlet kernel:

$$|\text{leakage}| \leq \frac{I_{h3}}{N \cdot |\Delta f / f_{bin}|} \quad (9)$$

For $N = 160$, $\Delta f = 141 - 3 \times 47 = 0$ Hz at 47 Hz (3rd harmonic falls exactly at $141 = 3 \times 47$ Hz), so leakage is exactly zero at 47 Hz as well. The harmonic regressors in (5) use the estimated frequency $\hat{f}$ from TR-20 to maintain orthogonality at any frequency in the 47–53 Hz range.

5 Integration with SAMBP Suite

Table 2: Position of TR-21 in the SAMBP 87L processing chain.

Stage	Module	Notes
1. Charging correction	<code>line_pi_section_model</code> (TR-15)	Fixed B_C ; superseded by TR-20
2. Frequency-adaptive B_C	<code>line_freq_adaptive</code> (TR-20)	Scales B_C by $\hat{f}/f_0$
3. Harmonic pre-filter	<code>line_harmonic_model</code> (TR-21, new)	Subtracts 3rd/5th/7th harmonics
4. LM estimator	<code>line_reduced_zone_model</code> (TR-11)	Unchanged; receives $\Delta i''$
5. Adaptive threshold	<code>line_adaptive_threshold</code> (TR-17)	$I_{thr} = \max(I_{base}, k_s I_{rest})$
6. Confidence gate	<code>line_confidence_gate</code> (TR-12)	$c = e^{-r_n}$; $c \geq 0.80$

The harmonic filter adds one stage between TR-20 and the LM estimator. It requires no modifications to any other module Blackburn and Domin [2006]. The H_{ratio} diagnostic can optionally be forwarded to the protection management system via GOOSE (IEC 61850) for network characterisation.

6 Computational Cost

For $N = 160$ samples and 6 regressors (3 harmonics $\times$ 2 basis functions), the least-squares solve requires:

- QR factorisation: $O(Nk) = O(960)$ multiply-accumulates
- Back-substitution: $O(k^2) = O(36)$ operations

On a typical IED DSP at 50 MFLOPS this takes < 0.02 ms, negligible compared to the 20 ms protection cycle. The harmonic pre-filter adds no latency to the trip path because it runs in the same cycle as the LM estimator IEC [2009].

7 Conclusions

1. **21/21 scenarios correct (100%)** across three studies (no-fault, internal fault, external fault) at seven harmonic levels (0–0.10 pu 3rd harmonic).
2. **Confidence fully restored:** TR-21 pre-filter achieves $c \geq 0.99$ in the no-fault study at all harmonic levels, vs $c \approx 0.37$ (below $c_{\min} = 0.80$) without the filter for $I_{h3} \geq 0.005$ pu.
3. **Fault detection unaffected:** I_{fund} estimates are invariant to harmonic injection due to fundamental–harmonic orthogonality over integer-cycle windows.
4. **Network-type diagnostic:** H_{ratio} cleanly separates SG-dominated (< 0.05), mixed, and IBR-dominated (> 0.20) regimes.
5. **Negligible computational cost:** < 0.02 ms per cycle (one QR factorisation of a 160×6 matrix).
6. **Frequency-adaptive harmonics:** regressors use $\hat{f}$ from TR-20, maintaining orthogonality across the 47–53 Hz range.

7.1 Future Work

1. **TR-22: Negative-sequence 87L for asymmetric IBR faults** — IBR current-limiting reduces positive-sequence contribution during single-phase faults; validate that negative-sequence differential (87LN) maintains sensitivity in compliance with IEC 60255-181.
2. **Inter-harmonic discrimination:** grid-interactive inverters can inject inter-harmonics (non-integer multiples of f_0). Extend the regressor matrix to include DFT-identified inter-harmonic bins.
3. **Hardware-in-the-loop at variable THD:** validate the pre-filter response during inverter fault ride-through transients using a real-time digital simulator with physical IBR hardware.
4. **Harmonic restraint element:** for very high THD networks (microgrids, EV charging hubs), consider adding an explicit harmonic restraint $I_{\text{harm,rms}} > I_{\text{harm,max}}$ as a second-level block, analogous to 2nd harmonic restraint in transformer differential.

References

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